Sample code for optimizing water retention curves

from **unsatfit** document: https://sekika.github.io/unsatfit/code-wrc.html

Easiest way to start learning how to use unsatfit is to run sample codes as instructed in this page. You can optimize parameters of WRF (water retention function) of various available models to measured data set. You can test with sample data provided in this page. SWRC Fit can also be used. For optimizing HCF (hydraulic conductivity function) also, please refer to this page.

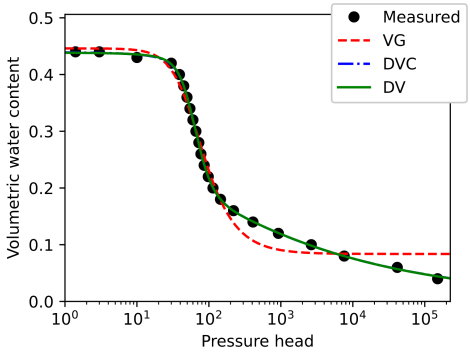
- Install Python 3 and unsatfit. Install pandas by python -m pip install pandas.
- Prepare datafile as csv (comma-separated values) format for (h, θ) data in the filename "swrc.csv" with a header of "h, theta". See sample data of Gilat loam.
- Download a sample code for the model that you select from below.
- Run the sample code at the same directory with the data file (swrc.csv). For running the code on Mac or unix-like system, edit the first line (shebang) and mark the file executable by <code>chmod +x VG.py</code>. For running on Windows, please refer to Python on Windows FAQ.
- Fitted parameters are shown at the standard output, where qs and qr means θ_s and θ_r respectively. Statistics are also shown.
- Note that the program is unit independent, meaning that the unit of the parameters depends on the unit of the input data. Unit of pressure head is assumed as cm for a (cm^{-1}) and h_b.
- Figure files are produced as svg files. For use in papers, pdf files can be produced as instructed in the sample code. To change the appearance of a figure, edit the figure settings in the sample code. For more figure options, see the source code of unsatfit.
- See Reading data from Excel if you wish.

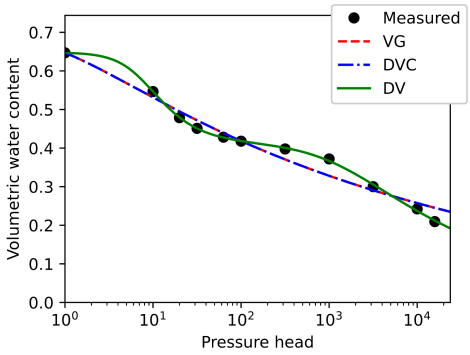
1.List of sample codes

- Brooks and Corey (BC) model
- van Genuchten (VG) model
- Kosugi (KO) model
- Fredlund und Xing (FX) model
- dual-BC (DB) model
- dual-BC-CH (DBC) model
- dual-VG (DV) model
- dual-VG-CH (DVC) model
- dual-KO (DK) model
- dual-KO-CH (DKC) model
- VG_1BC_2 (VB) model
- VG_1BC_2-CH (VBC) model
- KO_1BC_2 (KB) model
- KO_1BC_2-CH (KBC) model
- Peters (PE) model
- Fayer and Simmons (VGFS) model

2.Multiple curves

- Use sample code for multiple curves.
- It draws VG, dual-VG-CH (DVC) and dual-VG (DV) models in the same figure.
- It also shows the model with the smallest corrected AIC.
- Results with sample data of Gilat loam and Daisen Andisol (Fig. 9 in this paper) are shown below.





3.UNSODA data

Here are sample codes to draw multiple fitting curves for the UNSODA database and figures which were produced with the code. pdfgridcat was used for arranging the figures.

- VG series: VG, dual-VG-CH (DVC) and dual-VG (DV) models and figures.
- KO series: KO, KO_1BC_2-CH (KBC) and dual-KO (DK) models and figures.
- CH series: dual-BC-CH (DBC), VG_1BC_2-CH (VBC) and KO_1BC_2-CH (KBC) models and figures.
- dual series: dual-BC, dual-VG and dual-KO models and figures.

Method

- The code reads laboratory drying curves of UNSODA data from a JSON converted file.
- θ_r is fitted for VG and KO models, and $\theta_r = 0$ for other models.
- Upper limits are imposed for θ_s / max(θ) = 1.5 and n_1, n_2 = 8, along with a lower limit of $\sigma_2 = 0.2$.
- The generated PDF files are concatenated using this script.
- With the UNSODA viewer, you can select a specific soil sample and send data directly to SWRC Fit by clicking the link labeled "Fit with various water retention models".

4.Trimodal models

While trimodal water-retention functions provide the flexibility needed for media withclear triple porosity, they also introduce additional degrees of freedom and may lead to non-unique parameterizations when data coverage is limited or noisy. Please use these sample code with cautions. See Seki et al. (2026) for detail.

Here is sample code for trimodal models. The fixed parameters are θ_s = max(θ), θ_r = 0 and h_e (for VVP) = 6.3e6.

- tri-VG model
- $BC_1VG_2VG_3$ (BVV) model
- VVP model
- dual-VG and trimodal models

By running the final code on the IL sandstone sample dataset, you can generate this figure, which is identical to the upper panel of Figure 1 in Seki et al. (2026).

